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- (56)
- References Cited**

- (Continued)

- FOREIGN PATENT DOCUMENTS

- EP 3920399 A1 12/2021

- ## OTHER PUBLICATIONS

- Amoroso, Luca, et al., "Single Shot Transient Suppressor (SSTS) for High Current High Slew Rate Microprocessor", 14th Annual Applied Power Electronics Conference and Exposition, Dallas, TX, USA, Mar. 14-18, 1999, pp. 284-288.

- (Continued)

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Related U.S. Application Data

- (57) **ABSTRACT**

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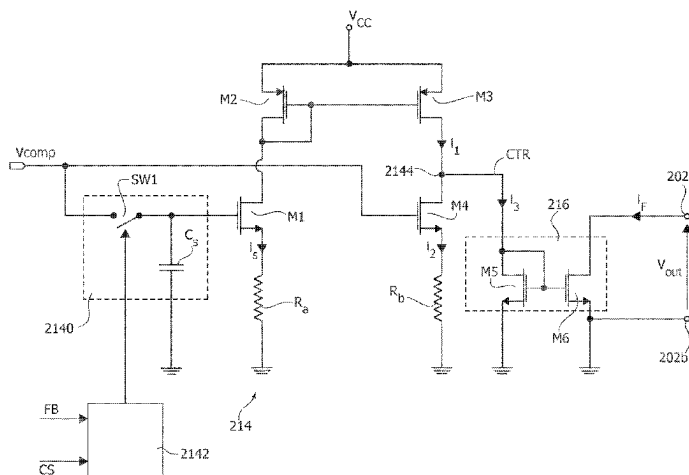
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(56)

References Cited

U.S. PATENT DOCUMENTS

9,178,420	B1	11/2015	Hawley	
9,257,905	B1	2/2016	Kotikalapoodi	
9,602,001	B1 *	3/2017	Hung	H02M 3/158
10,211,735	B1	2/2019	Chen et al.	
10,770,964	B1 *	9/2020	Su	H02M 1/08
10,797,597	B1 *	10/2020	Chang	H02M 3/158
11,394,301	B1	7/2022	Lisuwandi et al.	
2002/0030473	A1	3/2002	Utsunomiya	
2002/0125871	A1 *	9/2002	Groom	H02M 3/156
				323/284
2010/0201336	A1 *	8/2010	Chen	H02M 3/1588
				323/285
2011/0204859	A1 *	8/2011	Prodic	H02M 3/1588
				323/271
2013/0162235	A1	6/2013	Harada	
2016/0056714	A1	2/2016	Ek	
2016/0365789	A1 *	12/2016	Ejury	H02M 3/158
2017/0179824	A1 *	6/2017	Polhemus	G01R 19/0092
2019/0288603	A1	9/2019	Babazadeh et al.	

OTHER PUBLICATIONS

Wang, Pai-Yi, et al., "An Undershoot/Overshoot-Suppressed Current-Mode Buck Converter with Voltage-Setting Control for Type-II Compensator", IEEE Asian Solid-State Circuits Conference, Xiamen, Fujian, China, Nov. 9-11, 2015, 4 pages.

* cited by examiner

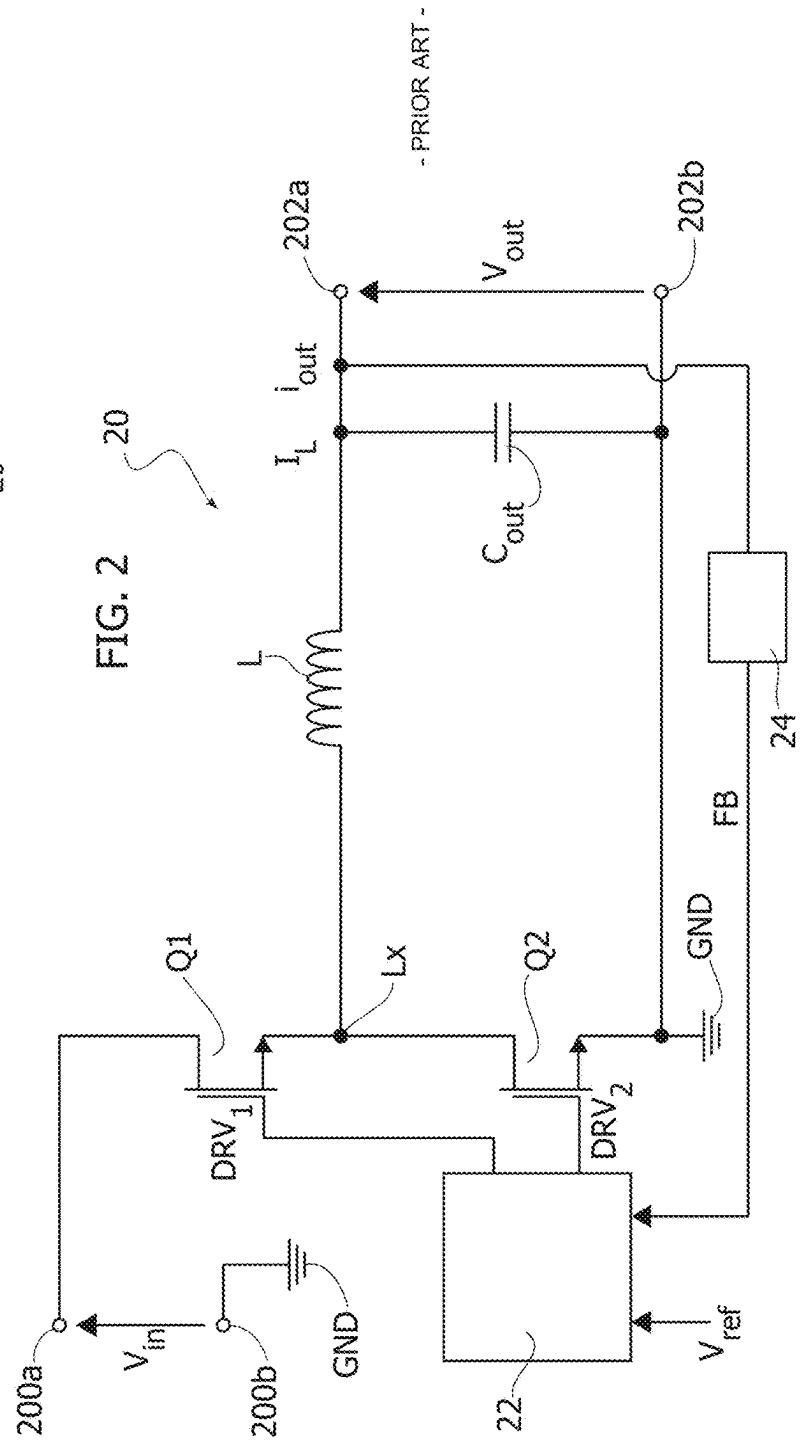
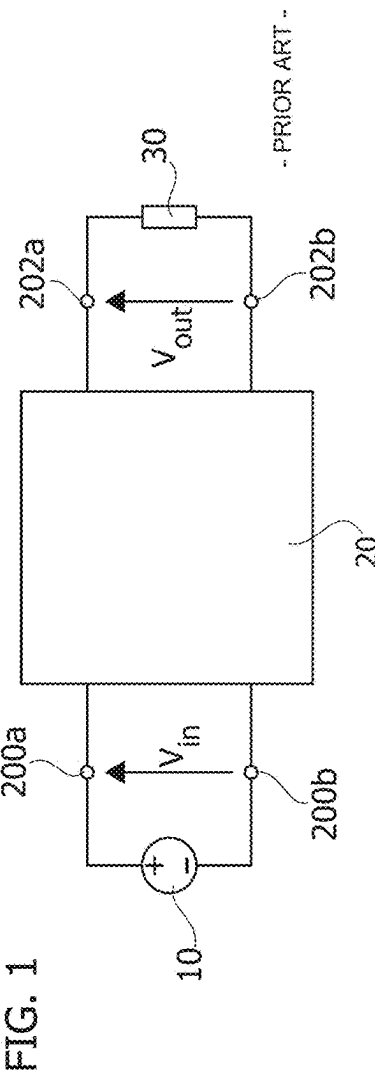
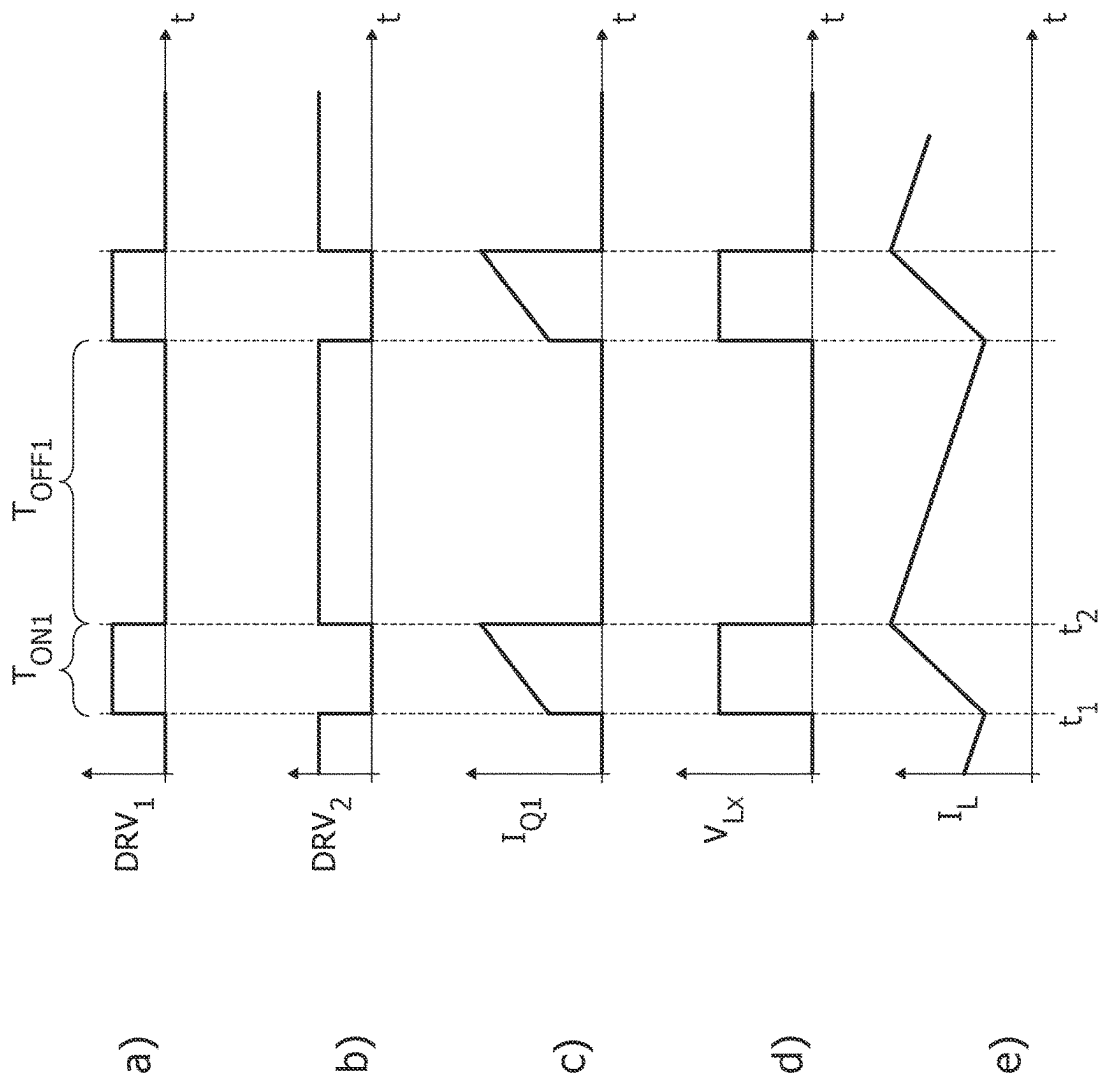
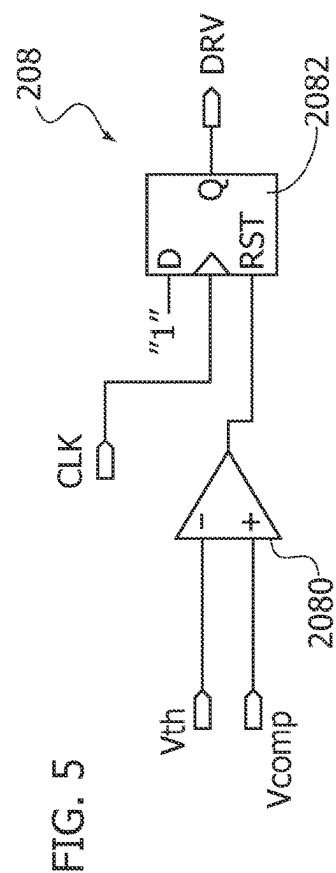
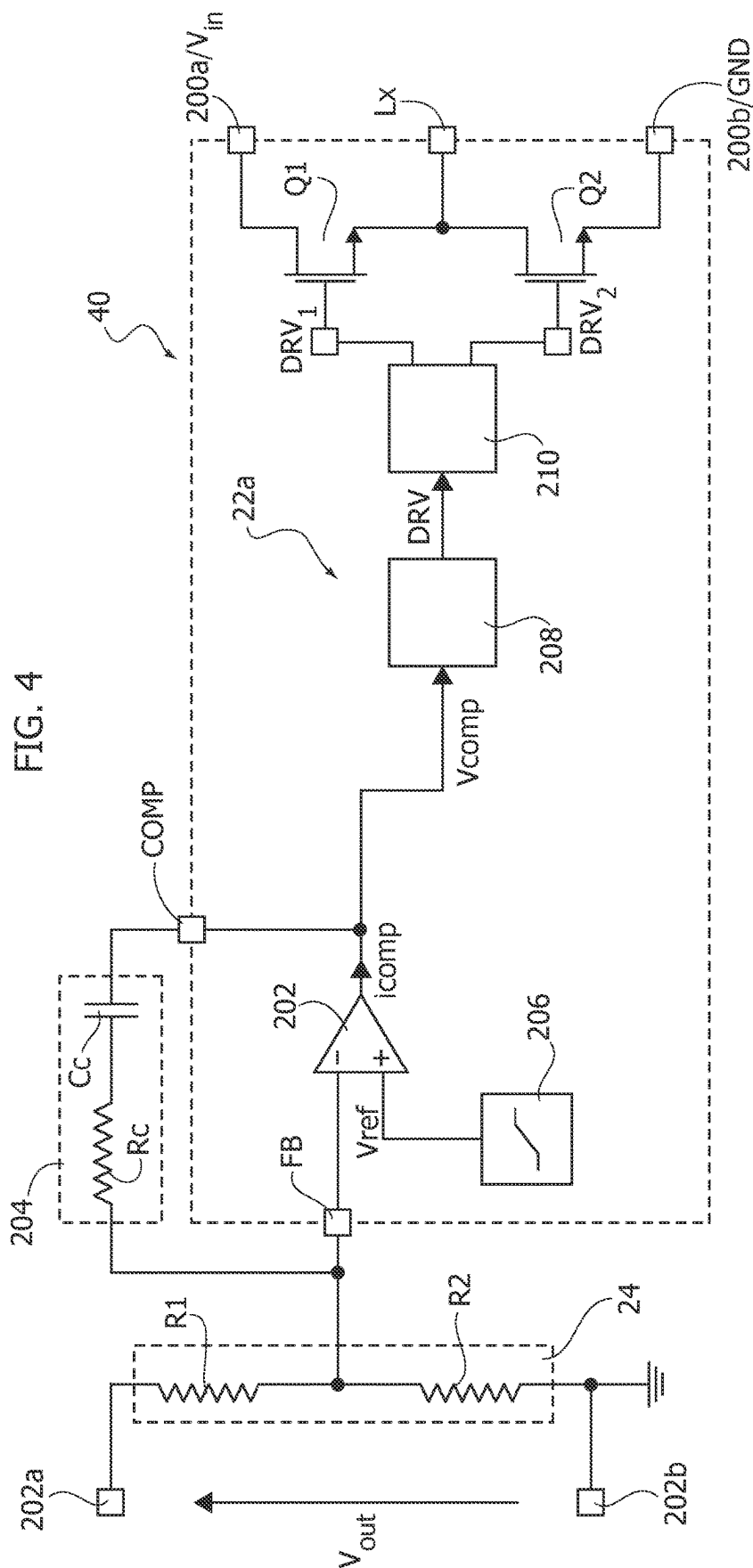
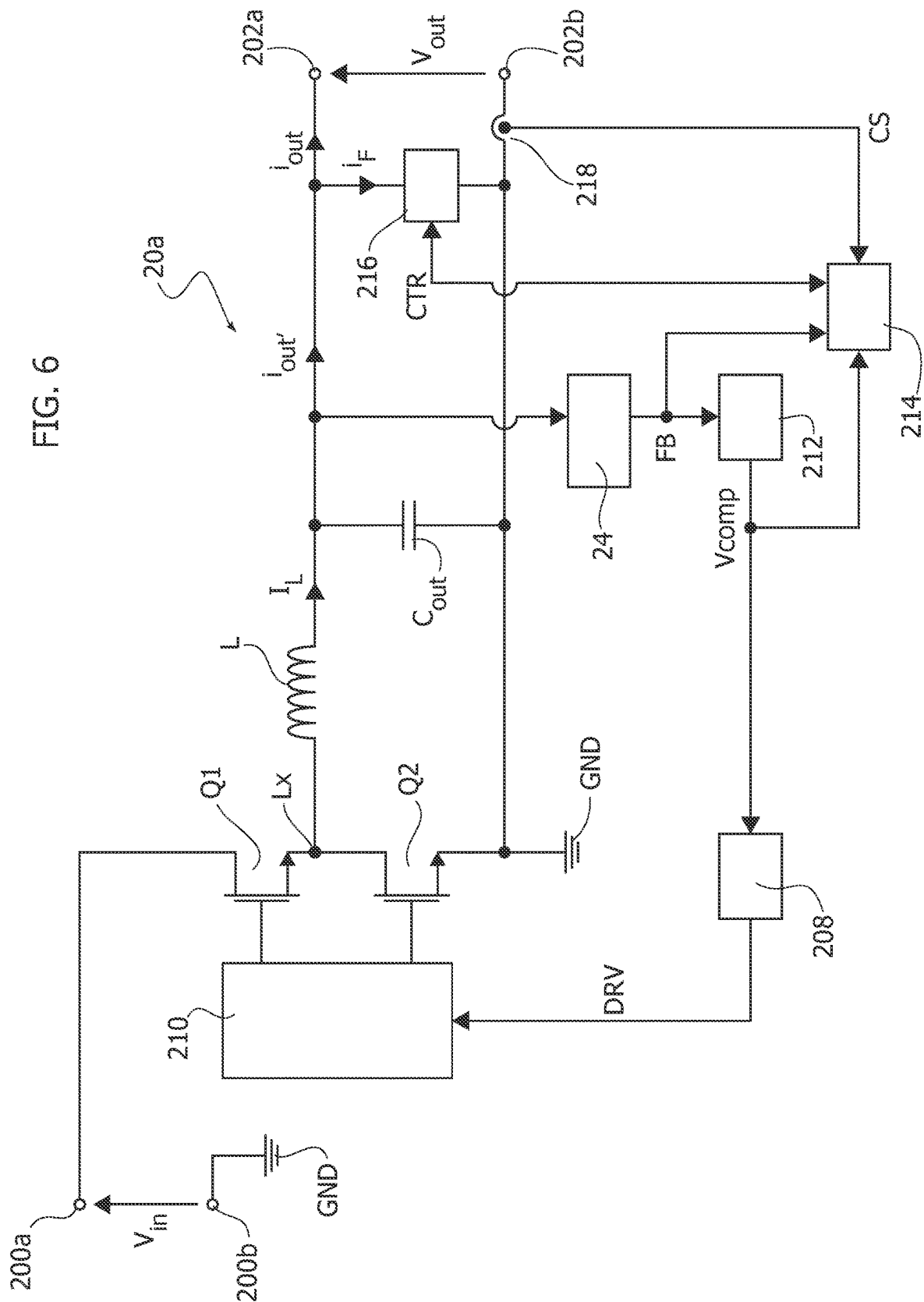


FIG. 3





6
G^x
E



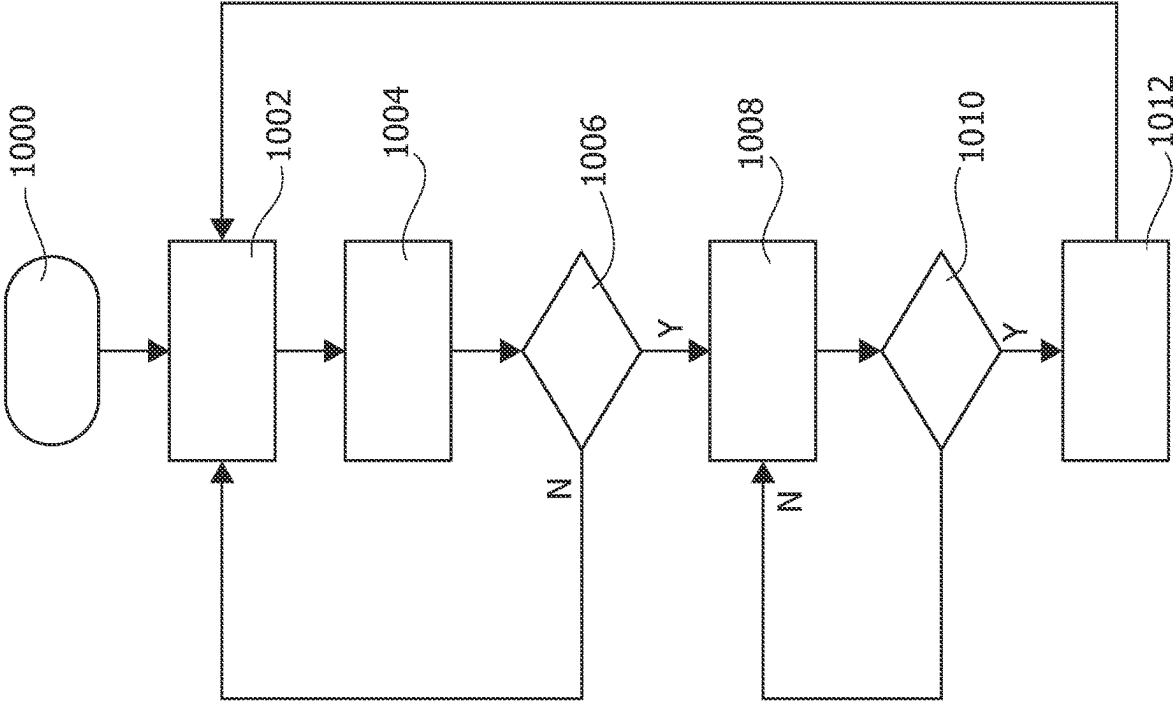
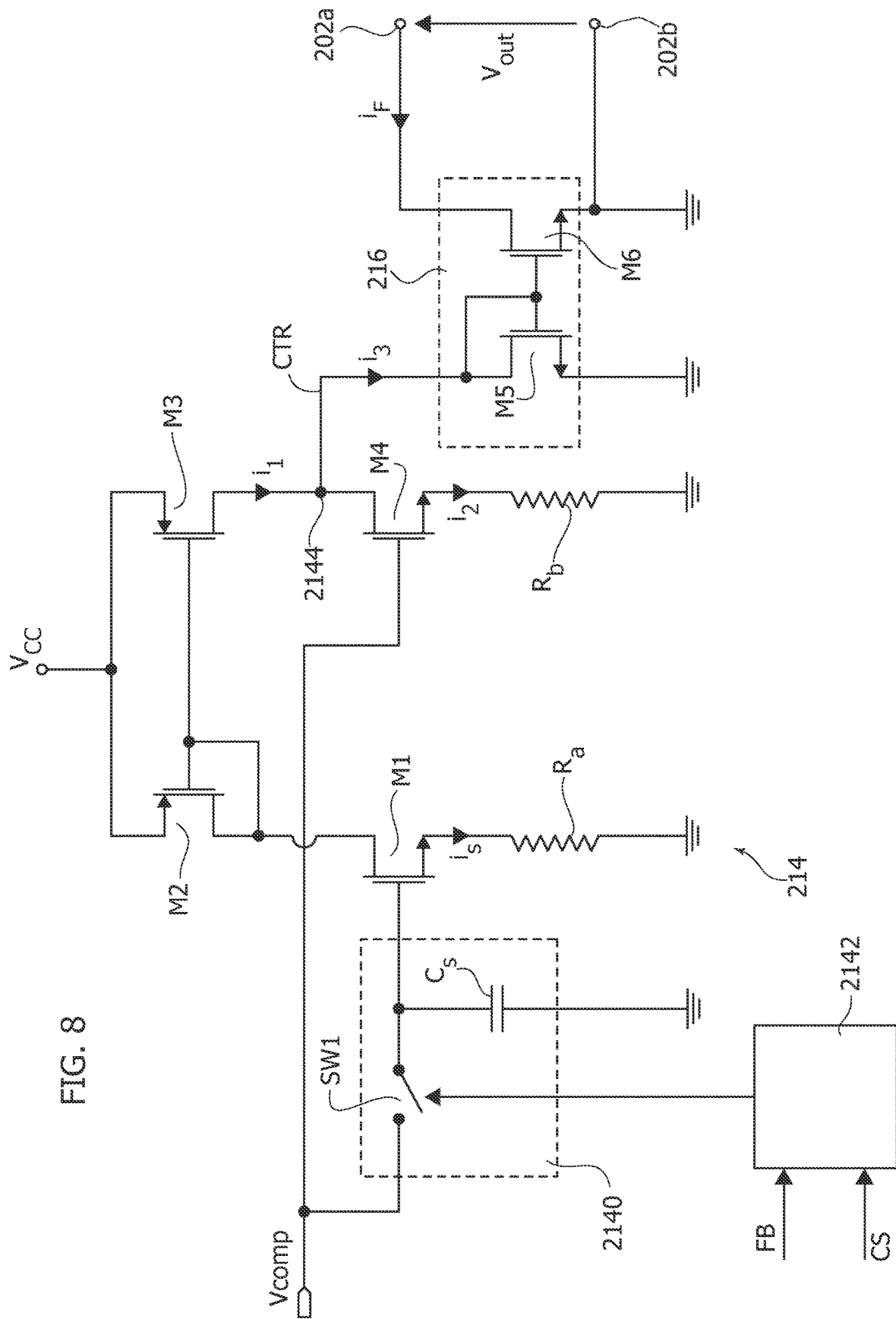
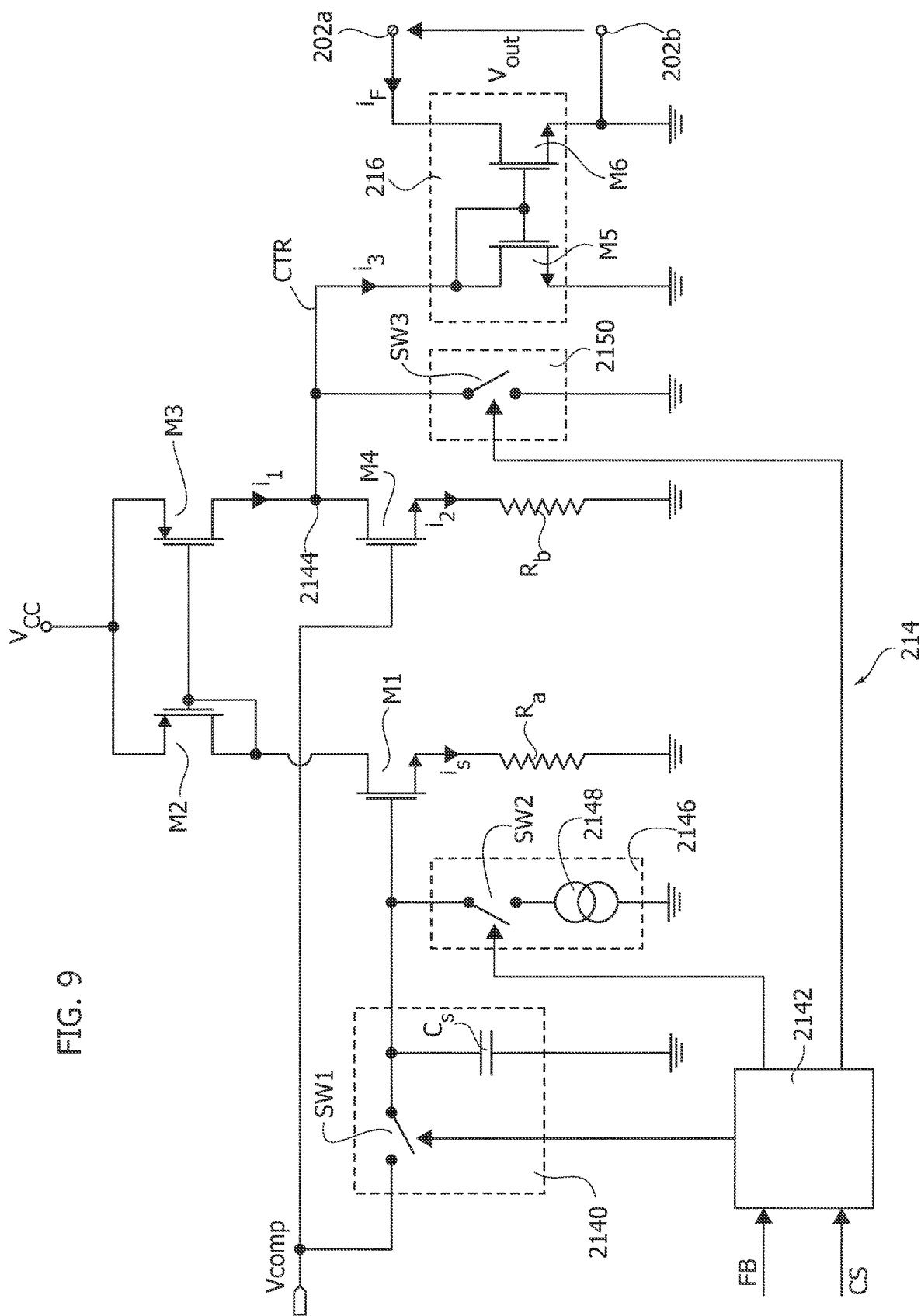
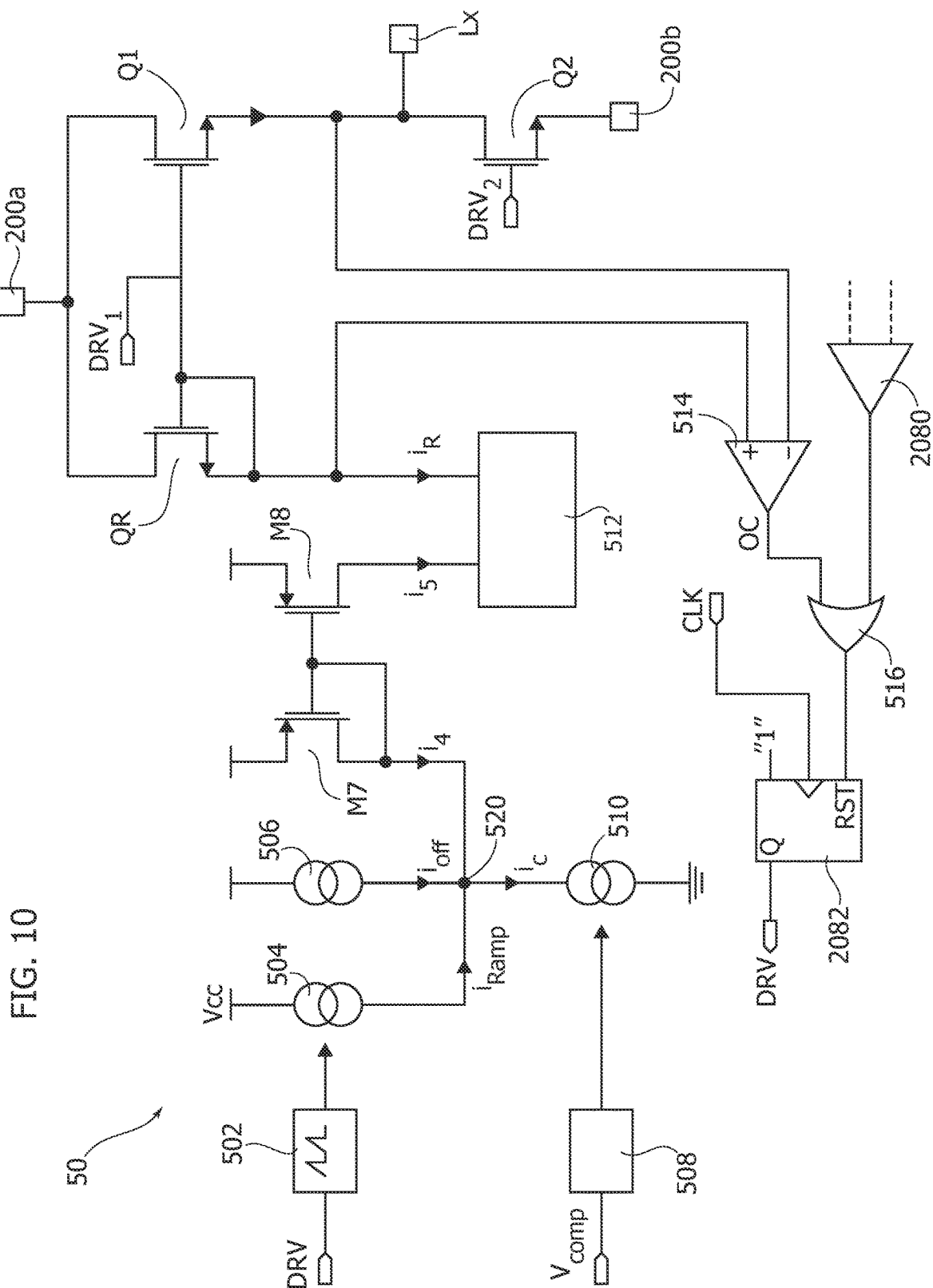
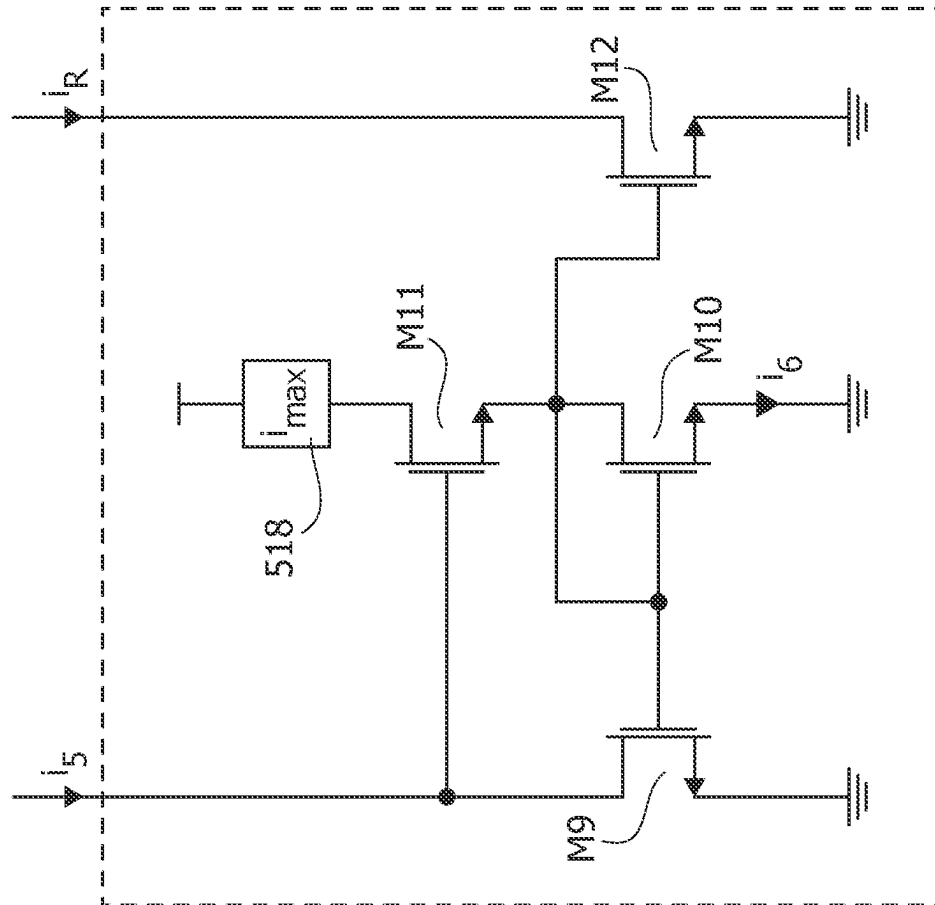


FIG. 7









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CONTROL CIRCUIT FOR REGULATING OUTPUT VOLTAGE OF A BUCK CONVERTER DURING NEGATIVE LOAD TRANSIENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/244,406, filed Apr. 29, 2021, which claims priority to Italian Application No. 102020000013231, filed on Jun. 4, 2020 which applications are hereby incorporated by their reference herein in their entirety.

TECHNICAL FIELD

Embodiments of the present description refer to a control device and method for a buck converter.

BACKGROUND

Power-supply circuits, such as AC/DC or DC/DC switched mode power supplies, are well known in the art. There exist many types of electronic converters, which are mainly divided into isolated and non-isolated converters. For instance, non-isolated electronic converters are the converters of the “buck”, “boost”, “buck-boost”, “Ćuk”, “SEPIC”, and “ZETA” type. Instead, isolated converters are, for instance, converters of the “flyback”, “forward”, “half-bridge”, and “full-bridge” type. Such types of converters are well known to the person skilled in the art.

FIG. 1 is a schematic illustration of a DC/DC electronic converter 20. In particular, a generic electronic converter 20 comprises two input terminals 200a and 200b for receiving a DC voltage V_{in} and two output terminals 202a and 202b for supplying a DC voltage V_{out} . For example, the input voltage V_i may be supplied by a DC voltage source 10, such as a battery, or may be obtained from an AC voltage by means of a rectifier circuit, such as a bridge rectifier, and possibly a filtering circuit. Instead, the output voltage V_{out} may be used to supply a load 30.

Voltage converters of a non-isolated step-down type are widely used, for example, in order to supply microcontrollers. The ease of use, simplicity, and excellent versatility in the various conditions of input and output voltage render the topology of a buck type one of the most widely used for this type of conversion.

FIG. 2 shows the circuit diagram of a typical buck converter 20. In particular, a buck converter 20 comprises two input terminals 200a and 200b for receiving a DC input voltage V_{in} and two output terminals 202a and 202b for supplying a regulated voltage V_{out} , where the output voltage is equal to or lower than the input voltage V_{in} .

In particular, typically, a buck converter 20 comprises two electronic switches Q1 and Q2 (with the current path thereof) connected (e.g. directly) in series between the input terminals 200a and 200b, wherein the intermediate node between the electronic switches Q1 and Q2 represents a switching node Lx. Specifically, the electronic switch Q1 is a high-side switch connected (e.g. directly) between the (positive) terminal 200a and the switching node Lx, and the electronic switch Q2 is a low-side switch connected (e.g. directly) between the switching node Lx and the (negative) terminal 200b, which often represents a ground GND. The (high-side) switch Q1 and the (low-side) switch Q2 hence

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represent a half-bridge configured to connect the switching node Lx to the terminal 200a (voltage V_{in}) or the terminal 200b (ground GND).

For example, the switches Q1 and/or Q2 are often transistors, such as Field-Effect Transistors (FETs), such as Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFETs), e.g. n-channel FET, such as NMOS. Frequently, the second electronic switch Q2 is also implemented just with a diode, where the anode is connected to the terminal 200b and the cathode is connected to the switching node Lx.

In the example considered, an inductance L, such as an inductor, is connected (e.g. directly) between the switching node Lx and the (positive) output terminal 202a. Instead, the (negative) output terminal 202b is connected (e.g. directly) to the (negative) input terminal 200b.

In the example considered, to stabilise the output voltage V_{out} , the converter 20 typically comprises a capacitor C_{out} connected (e.g., directly) between the output terminals 202a and 202b.

In this context, FIG. 3 shows some waveforms of the signals of such an electronic converter, where:

FIG. 3a shows the signal DRV_1 for switching the electronic switch Q1;

FIG. 3b shows the signal DRV_2 for switching the second electronic switch Q2;

FIG. 3c shows the current I_{Q1} that traverses the electronic switch Q1;

FIG. 3d shows the voltage V_{Lx} at the switching node Lx (i.e., the voltage at the second switch Q2); and

FIG. 3e shows the current I_L that traverses the inductor L.

In particular, when the electronic switch Q1 is closed at an instant t_1 (ON state), the current I_L in the inductor L increase (substantially) linearly. The electronic switch Q2 is at the same time opened. Instead, when the electronic switch Q1 is opened after an interval T_{ON1} at an instant t_2 (OFF state), the electronic switch Q2 is closed, and the current I_L decreases (substantially) linearly. Finally, the switch Q1 is closed again after an interval T_{OFF1} . In the example considered, the switch Q2 (or a similar diode) is hence closed when the switch Q1 is open, and vice versa.

The current I_L can thus be used to charge the capacitor C_{out} , which supplies the voltage V_{out} at the terminals 202a and 202b.

In general, the electronic converter 20 hence comprises a control circuit 22 configured to drive the switching of the switch Q1, and possibly of the switch Q2, for repeating the intervals T_{ON1} and T_{OFF1} periodically. For example, typically the buck converter 20 comprises also a feedback circuit 24, such as a voltage divider, configured to generate a feedback signal FB indicative of (and preferably proportional to) the output voltage V_{out} , and the control circuit 22 is configured to generate the drive signals DRV_1 and DRV_2 by comparing the feedback signal FB with a reference signal, such as a reference voltage V_{ref} .

A significant number of driving schemes are known for the switch Q1, and possibly for the switch Q2. These solutions have in common the possibility of regulating the output voltage V_{out} by regulating the duration of the interval T_{ON1} and/or the interval T_{OFF1} . For instance, in many applications, the control circuit 22 generates a driving signal DRV_1 for the switch Q1 (and possibly a driving signal DRV_2 for the switch Q2), where the driving signal DRV_1 is a Pulse-Width Modulation (PWM) signal, i.e., the duration of the switching interval $T_{SW1} = T_{ON1} + T_{OFF1}$ is constant, but the duty cycle T_{ON1}/T_{SW1} may be variable. In this case, the control circuit 14 typically implements a regulator circuit

having a Proportional (P) and/or Integral (I) component, wherein the regulator circuit is configured to vary the duty cycle of the signal DRV_1 in order to obtain a required output voltage V_{out} . In this case, the various operating modes of the converter (Continuous-Conduction Mode, CCM; Discontinuous-Conduction Mode, DCM; Transition Mode, TM) are well known in the technical field.

One of the most important parameters in a buck converter is the load regulation, which is the capability of the circuit to keep the output voltage V_{out} stable in response to changing load conditions, which also implies a varying output current i_{out} . When the output current i_{out} changes in the time, overshoots and undershoots can be observed in the output voltage V_{out} as a function of the ratio $\pm \Delta i_{out} / \Delta T$, wherein Δi_{out} represents the variation of the current i_{out} in a given time interval ΔT . In fact, when the output current i_{out} changes, the current I_L supplied by the inductor L may be too high or too low, thereby creating a variation of the voltage V_{out} at the capacitor C_{out} .

For example, as mentioned before, such a buck converter may be used to supply a microcontroller, which may also be configured to drive other loads. In this case, significant load transitions may occur, such as negative load transient of 1 A/1 μ s. However, similar load transitions may occur each time the load **30** is a switchable load, which e.g. may comprise one or more loads selectively connected to the output voltage V_{out} . For example, assuming a buck converter **20** configured to supply an output voltage V_{out} in a range between 1V and 1.31 V, which e.g. may be settable by adjusting the reference voltage of the control circuit **22**, and a maximum overshoot of 6%, the circuit **20** has to be able to maintain the overshoot in the voltage V_{out} below 60 mV for the negative load transient of 1 A/1 μ s.

SUMMARY

Considering the foregoing, it is therefore an object of various embodiments to provide a control device for a buck converter able to better regulate the output voltage in response to load transients, in particular in response to a negative current transient.

According to one or more embodiments, one or more of the above objects are achieved by a driver circuit for a buck converter having the distinctive elements set forth specifically in the ensuing claims. Embodiments moreover concern a related integrated circuit, electronic buck converter and method.

The claims form an integral part of the technical teaching of the description provided herein.

As mentioned before, various embodiments of the present disclosure relate to a control circuit for a buck converter configured to provide via two output terminals an output voltage and an output current. For example, such a control circuit may be implemented in an integrated circuit.

In various embodiments, the control circuit comprises a first terminal configured to be connected to an electronic switch of the buck converter, a second terminal configured to receive a feedback signal indicative of the output voltage, and two terminals configured to be connected to the two output terminals of the buck converter.

In various embodiments, the control circuit comprises an error amplifier configured to generate an error signal as a function of the feedback signal and a reference signal. For example, in various embodiments, the error amplifier is a regulator having an integral and/or a proportional component.

In various embodiments, the control circuit comprises a pulse generator circuit configured to generate a pulsed signal having switching cycles where the pulsed signal is set to high for a first duration and low for a second duration, wherein the pulse generator circuit is configured to vary the first and/or the second duration as a function of the error signal, and a driver circuit configured to generate a drive signal at the first terminal as a function of the pulsed signal. For example, in various embodiments, the pulse generator circuit is a pulse width modulator, wherein the pulsed signal is a pulse width modulated signal having a constant frequency and a duty cycle determined as a function of the error signal.

In various embodiments, the control circuit comprises also a variable load connected between the two terminals, wherein the variable load is configured to absorb a current determined as a function of a control signal.

In this case, a detector circuit may be configured to generate the control signal by monitoring a first signal indicative of the output current, and a second signal indicative of a negative transient of the output current. For example, the variable load may comprise a current mirror receiving at an input the control signal, and wherein an output of the current mirror is connected between the two terminals. For example, in case the error amplifier is a regulator having an integral component, the first signal may correspond to the error signal. Conversely, the detector circuit may comprise a transient detection circuit configured to determine the second signal by verifying whether the first signal indicates that the output current decreases more than a given amount, and/or verifying whether the feedback signal indicates that the output voltage increases more than a given amount, and/or verifying whether the feedback signal indicates that the output voltage exceeds a given maximum threshold value.

In various embodiments, the detector circuit is thus configured to verify whether the second signal indicates a negative transient of the output current. Specifically, when the second signal does not indicate a negative transient of the output current, the detector circuit stores the monitored first signal. Conversely, when the second signal indicates a negative transient of the output current, the detector circuit generates the control signal as a function of the difference between the stored first signal and the monitored first signal.

In various embodiments, the detector circuit may also verify whether the second signal indicates that the output current is again stable, and when the second signal indicates that the output current is again stable, the detector circuit may gradually reduce the control signal, whereby the variable load absorbs gradually less current.

In various embodiments, the control circuit may also implement an over-current protection. For this purpose, the control circuit may comprise a current ramp generator configured to generate a current ramp signal when the pulsed signal is set to high, a reference electronic switch, wherein the current ramp signal flows through the reference electronic switch when the pulsed signal is set to high, and a comparator circuit configured to compare the voltage at the reference electronic switch with the voltage at the electronic switch of the buck converter, wherein the comparator circuit is configured to set the pulsed signal to low, when the voltage at the electronic switch of the buck converter exceeds the voltage at the reference electronic switch.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of the present disclosure will now be described with reference to the annexed plates of drawings, which are provided purely to way of non-limiting example and in which:

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The features and advantages of the present invention will become apparent from the following detailed description of practical embodiments thereof, shown by way of non-limiting example in the accompanying drawings, in which:

FIG. 1 shows a typical application of an electronic converter;

FIG. 2 shows an example of a buck converter;

FIG. 3 shows exemplary waveforms of the converter of FIG. 2;

FIG. 4 shows an embodiment of a control circuit for a buck converter;

FIG. 5 shows an embodiment of a pulse generator circuit for the control circuit of FIG. 4;

FIG. 6 shows an embodiment of a buck converter comprising a detector circuit and a variable load;

FIG. 7 shows an embodiment of the operation of the buck converter of FIG. 6;

FIG. 8 shows a first embodiment the detector circuit and the variable load of FIG. 6;

FIG. 9 shows a second embodiment the detector circuit and the variable load of FIG. 6; and

FIGS. 10 and 11 show embodiments of a current limiter for the high side switch of the buck converter of FIG. 6.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

In the ensuing description, various specific details are illustrated aimed at enabling an in-depth understanding of the embodiments. The embodiments may be provided without one or more of the specific details, or with other methods, components, materials, etc. In other cases, known structures, materials, or operations are not shown or described in detail so that various aspects of the embodiments will not be obscured.

Reference to “an embodiment” or “one embodiment” in the framework of this description is meant to indicate that a particular configuration, structure, or characteristic described in relation to the embodiment is comprised in at least one embodiment. Hence, phrases such as “in an embodiment”, “in one embodiment”, or the like that may be present in various points of this description do not necessarily refer to one and the same embodiment. Moreover, particular conformations, structures, or characteristics may be combined in any adequate way in one or more embodiments.

The references used herein are only provided for convenience and hence do not define the sphere of protection or the scope of the embodiments.

In FIGS. 4 to 11 described below, parts, elements or components that have already been described with reference to FIGS. 1 to 3 are designated by the same references used previously in these figures. The description of these elements has already been made and will not be repeated in what follows in order not to burden the present detailed description.

As explained in the foregoing, various embodiments of the present description relate to a control circuit 22a for a buck converter.

FIG. 4 shows a first embodiment of a control circuit 22a for a buck converter. For a general description of a buck converter 20, reference may be made to the description of FIGS. 1 to 3. Specifically, as described in the foregoing, a buck converter 20 comprises:

two input terminals 200a and 200b for receiving an input voltage V_{in} ;

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two output terminals 202a and 202b for providing an output voltage V_{out} ;

two electronic switches Q1 and Q2, such as NMOS (wherein the electronic switch Q2 may also be a diode), connected in series between the terminals 200a and 200b, wherein the intermediate node represents a switching node Lx;

an inductance L, such as an inductor, connected between the switching node Lx and the output terminal 202a;

a capacitor Cout connected between the two output terminals;

a feedback circuit 24, corresponding to a voltage measurement circuit, configured to generate a signal indicative of the output voltage V_{out} ; and

a control circuit 22a configured to generate the drive signals for the electronic switch Q1 and optionally the electronic switch Q2 as a function of the feedback signal FB.

As shown in FIG. 4, in various embodiments, such a control circuit 22a may be implemented in an integrated circuit 40, e.g. the integrated circuit 40 may comprise:

a pad (of a respective die) or pin (of a respective integrated circuit package) configured to be connected to the feedback circuit 24, such as a voltage divider comprising two or more resistors R1 and R2 connected in series between the output terminals 202a and 202b, whereby the feedback signal FB is a voltage proportion to the output voltage V_{out} ;

a pad/pin for providing the drive signal DRV_1 to the electronic switch Q1 and optionally a pad/pin for providing the drive signal DRV_2 to the electronic switch Q2, e.g. to the gate terminals of respective FETs; and

the control circuit 22a.

As shown in FIG. 3, in various embodiments, the integrated circuit 40 may also comprise the electronic switch Q1 and optionally the electronic switch (or diode) Q2. For example, in this case, the two pads/pins for providing the drive signal DRV_1 and DRV_2 may be omitted and the integrated circuit may comprise pads/pins for connection to the terminals 200a, 200b and Lx. In various embodiments, the integrated circuit may in any case be connected to the terminals 200a/200b insofar as these terminals may be used to provide the power supply to the control circuit 22a. Alternatively, the power supply for the control circuit 22a may be obtained, e.g., from the input voltage V_i (terminals 200a/200b) or the output voltage V_{out} (terminals 202a/202b).

In various embodiments, the integrated circuit 40 may also comprise the feedback circuit/voltage measurement circuit 24. In this case, the feedback pad/pin may be omitted and the integrated circuit 40 may comprise two pins/pads for connection to the terminals 202a and 202b. As mentioned before, the integrated circuit 40 may already be connected to the terminal 202b and accordingly the pin/pad for the terminal 202b may be omitted.

Thus, irrespective of the integration of the various blocks in the integrated circuit 40, the control circuit 22a comprises:

a terminal/node for receiving the feedback signal FB indicative of (and preferably proportional to) the output voltage V_{out} ; and

a terminal/node for providing the drive signal DRV_1 and optionally a terminal/node for providing the drive signal DRV_2 .

Specifically, in the embodiment considered, the control circuit 22a comprises an error amplifier configured to gen-

erate an error signal V_{comp} , by comparing the feedback signal FB with a reference signal, such as a reference voltage V_{ref} .

For example, in the embodiment considered, the error amplifier is implemented with an operational amplifier **202** and a compensation/feedback network **204** associated with the operational amplifier **202**. For example, in the embodiment considered, the feedback signal FB is connected to the inverting/negative input of the operational amplifier **202** and the reference voltage V_{ref} is connected to the non-inverting/positive input terminal of the operational amplifier **202**.

In various embodiments, the compensation network **204** is connected between the output of the operational amplifier **202** and the feedback terminal (e.g. the inverting input of the operational amplifier **202**) and/or ground GND, which as mentioned before may correspond to the terminal **200b** and/or **202b**.

For example, in the embodiment considered, the compensation network **204** comprises at least one capacitor Cc (integral component) and/or at least one resistor Rc (proportional component). For example, in the embodiment considered, a resistor Rc and a capacitor Cc are connected in series between the feedback terminal and the output of the operational amplifier **202**. Specifically, in the embodiment considered, the operational amplifier **202** provides a current i_{comp} as a function of the difference between the reference voltage V_{ref} and the feedback signal FB, and the compensation network **204**, e.g. via the resistor Rc and/or the capacitor Cc, is configured to convert the current i_{comp} into the error signal/voltage V_{comp} .

In general, the compensation network **204** may be integrated in or may be external to the integrated circuit **40**. For example, in the embodiment considered, the integrated circuit **40** comprises a pin/pad COMP connected to the output of the operational amplifier **202**, and the compensation network **204** may be connected (e.g. externally) between the pin/pad COMP and ground GND.

Accordingly, in various embodiments, the error amplifier **202/204** may be configured as regulator comprising an I (Integral) and/or a P (Proportional) component.

In the embodiment considered, the error signal V_{comp} is provided to a pulse generator circuit **208**, such as a PWM generator circuit, configured to generate a binary/pulsed signal DRV which is alternatively set to a first logic level (e.g. high) and a second logic level (e.g. low) for respective durations T_1 and T_2 . Specifically, the pulse generator circuit **208** is configured to vary at least one of the durations T_1 and T_2 as a function of the error signal V_{comp} .

In the embodiment considered, the binary/pulsed signal DRV is provided to a driver circuit **210** configured to generate the drive signal DRV_1 and DRV_2 as a function of the drive signal DRV. For example, in various embodiments, the driver circuit **210** may be configured to:

set the signal DRV_1 to high when the signal DRV has the first logic level and to low when the signal DRV has the second logic level; and

optionally set the signal DRV_2 to high when the signal DRV has the second logic level and to low when the signal DRV has the first logic level.

Accordingly, in various embodiments the switch-on and switch-off durations T_{ON1} and T_{OFF1} correspond to the durations T_1 and T_2 , respectively. For example, the logic levels of the signal DRV_1 may correspond to the logic levels of the signal DRV, and optionally the logic levels of the signal DRV_2 may correspond to the inverted version of the logic levels of the signal DRV.

Accordingly, essentially, the feedback loop via the feedback circuit **24** and the error amplifier **202/204** varies the error signal V_{comp} , which in turn is used by the pulse generator circuit **208** to drive via the driver circuit **210** the electronic switches Q1 and Q2, thereby regulating the output voltage V_{out} to a requested value, which may be determined, e.g., as a function of the scaling factor of the feedback circuit **24** and the reference voltage V_{ref} .

In various embodiments, the reference voltage V_{ref} may also be provided by a soft-start circuit **206** configured to increase, in response to a power-on of the control circuit **22a**, the reference voltage V_{ref} from a minimum value (e.g. 0 V) to a maximum value (corresponding to the nominal value of the reference voltage V_{ref}).

FIG. 5 shows a possible embodiment of the pulse generator circuit **208**. For example, in the embodiment considered, the pulse generator circuit **208** is a fixed frequency PWM generator circuit, i.e. the signal DRV has a constant switching frequency ($T_{SW}=T_1+T_2$) and a variable duty cycle ($D=T_1/T_{SW}$).

For example, in the embodiment considered, the error signal V_{comp} is provided to an input (e.g. the positive input terminal) of a comparator **2080** configured to determine whether the error signal V_{comp} is smaller or greater than a threshold value V_{th} .

Specifically, in the embodiment considered, the signal at the output of the comparator **2080** is fed to a flip-flop **2082** configured to set the signal DRV to:

high with the rising edge (or alternatively the falling edge) of a clock signal CLK; and
low when the error signal V_{comp} reaches or exceeds the threshold value V_{th} .

For example, in the embodiment considered, the flip-flop **2082** is a D type flip-flop receiving:

at a clock input the clock signal CLK;
at a data input the logic value high ("1"); and
at a reset input the signal provided by the comparator **2080**.

Thus, in the embodiment considered, the signal DRV is set to high with a fixed frequency (determined by the clock signal CLK) and then set to low when the signal V_{comp} reaches or exceeds the threshold value V_{th} .

Generally, the embodiment shown in FIG. 5 may also be adapted to support an over-voltage protection and/or burst mode. Specifically, by using an additional comparator configured to determine whether the feedback signal FB exceeds a given maximum threshold value, the flip-flop **2082** may receive a masked version of the clock signal CLK, whereby:

when the feedback signal FB is smaller than the maximum threshold, the flip-flop **2082** is periodically set via the clock signal CLK; and
when the feedback signal FB is greater than the maximum threshold, the clock signal CLK is masked and the flip-flop **2082** remains reset.

As mentioned before, various embodiments of the present disclosure relate to solutions, which permit to limit the overshoot of the output voltage V_{out} when the output current i_{out} varies rapidly.

For example, in the control circuit **22a** shown in FIG. 4 there will be intrinsic delays (e.g. due to the operational amplifier **202**, the compensation network and the pulse generator circuit **208**, e.g. the comparator **2080**) in the regulation loop, which usually cannot ensure that the output voltage V_{out} does not increase significantly, e.g. because

there may still be stored energy in the inductance L or the error signal V_{comp} will not follow immediately the load variation.

FIG. 6 shows an embodiment of a modified buck converter **20a**, and in particular a modified control circuit **22a**. Specifically, also in this case, the control circuit **22a** comprises:

- a terminal configured to be connected to a feedback circuit **24** configured to generate a feedback FB as a function of the output voltage V_{out} ;
- an error amplifier **212** configured to generate an error signal V_{comp} as a function of the feedback FB , wherein the error amplifier **212** may comprise the operational amplifier **202**, the compensation network **204** and optionally the soft-start circuit **206** of FIG. 4;
- a pulse generator circuit **208** configured to generate a pulsed signal DRV ; and
- a driver circuit **210** configured to generate the drive signal DRV_1 and optionally the drive signal DRV_2 as a function of the signal DRV .

Accordingly, these blocks are configured to regulate during normal operation the output voltage V_{out} to a given requested value.

In the embodiment considered, the control circuit **22a** comprises a variable load **216** configured to absorb a current i_F as a function of the control signal CTR and a detector circuit **214** configured to generate the control signal CTR (at least) as a function of a signal being indicative of the output current i_{out} .

FIG. 7 shows an embodiment of the operation of the detector circuit **214**. In the embodiment considered, after a start step **1000**, the detector circuit **214** monitors at a step **1002** the signal being indicative of the output current i_{out} provided via the output terminals **202a** and **202b**. For example, the signal being indicative of the output current i_{out} may be a current sense signal CS provided by a current sensor **218**, such as a shunt resistor, connected in series with the output terminals **202a** and **202b**. Alternatively, in case the compensation network **204** comprises a capacitor C_c , whereby the error signal V_{comp} is proportional to (or at least indicative of) the output current i_{out} , the signal being indicative of the output current i_{out} may correspond to the error signal V_{comp} .

At a following step **1004**, the detector circuit **214** stores the signal being indicative of the output current i_{out} . For example, the detector circuit **214** may store the signal being indicative of the output current i_{out} periodically or in response to given events.

At a following step **1006**, the detector circuit **214** verifies whether a negative transition of the output current occurs. For example, for this purpose, the detector circuit **214** may verify at least one of the following conditions:

- verify whether the signal being indicative of the output current i_{out} (e.g. the signal V_{comp}) indicates that the output current i_{out} decreases more than a given amount (relative variation), e.g. by comparing the current value of the signal being indicative of the output current i_{out} with a respective stored value;
- verify whether the feedback signal FB indicates that the output voltage V_{out} increases more than a given amount (relative variation), e.g. by comparing the current value of the feedback signal FB with a respective stored value; or
- verify whether the feedback signal FB indicates that the output voltage V_{out} exceeds a given maximum threshold value (absolute value).

In case no negative transient of the output current is detected (output “N” of the verification step **1006**), the detector circuit **214** may return to the step **1002** for performing a new monitoring operation.

Conversely, in case a negative transition of the output current is detected (output “Y” of the verification step **1006**), the detector circuit **214** proceeds to a step **1008**, where the detector circuit **214** generates the control signal CTR . Specifically, in various embodiments, the control signal CTR is indicative of the difference between the stored signal and the current value of the signal being indicative of the output current i_{out} . Accordingly, when the converter **20a** provides a given output current $i_{out,1}$ and the output current is reduced to a value $i_{out,2}$, the detector circuit **214** drives the variable load **216** via the signal CTR in order to absorb a current $i_F = i_{out,1} - i_{out,2}$, whereby the total current $i_{out}' = i_F + i_{out,1}$ provided by the capacitor C_{out} remains constant.

The step **1006** is thus purely optional, because when the monitored signal corresponds to the stored signal (static condition), also the signal CTR would indicate that the difference is zero and the variable load **216** would absorb a current $i_F = 0$.

In various embodiments, the detector circuit **214** proceeds then to a verification step **1010**, where the detector circuit **214** verifies whether the output current i_{out} remains stable. For example, for this purpose, the detector circuit **214** may verify at least one of the following conditions:

- verify whether the signal being indicative of the output current i_{out} (e.g. the signal V_{comp}) indicates that the output current i_{out} varies less than a given amount (relative variation), e.g. by comparing the current value of the signal being indicative of the output current i_{out} with a respective stored value;
- verify whether the feedback signal FB indicates that the output voltage V_{out} varies less than a given amount (relative variation), e.g. by comparing the current value of the feedback signal FB with a respective stored value; or
- verify whether the feedback signal FB indicates that the output voltage V_{out} is below a given maximum threshold value (absolute value).

In case the output current i_{out} is not stable (output “N” of the verification step **1010**), the detector circuit returns to the step **1008**.

Conversely, in case the output current i_{out} is stable (output “Y” of the verification step **1010**), the detector circuit **214** may reduce (preferably gradually and slowly) the control signal CTR , e.g. by reducing the stored value, whereby the variable load **216** absorbs less current i_F and the feedback loop regulated the output voltage V_{out} as a function of the new load condition. However, insofar as the detector circuit **214** is configured to vary the control signal CTR with a time constant being greater than the time constant of the feedback loop, the feedback loop is able to follow the load variation without overshoots in the output voltage V_{out} .

Accordingly, once the detector circuit **214** has reduced the current i_F to zero via the control signal CTR , the detector circuit **214** may return to the step **1002** for detecting a following load transition.

Accordingly, in various embodiments, the detector circuit **214** is configured to track the output current load and sample its value. This sampled value is used to apply an internal current load i_F to the output terminals **202a** and **202b** in order to replace the reduction of the external load current. Next, the detector circuit **214** may decrease the internal current load i_F with a controlled slope minimizing overshoot.

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FIG. 8 shows a possible embodiment of the detector circuit **214** and the variable load **216**. Specifically, in the embodiment considered, the signal being indicative of the output current i_{out} corresponds to the error signal V_{comp} . However also the current sense signal CS may be used.

In the embodiment considered, the error signal V_{comp} is provided to an analog sample-and-hold circuit **2140**. For example, such a sample-and-hold circuit **2140** may be implemented with a storage capacitor C_S and an electronic switch SW1 configured to connect the storage capacitor C_S to the error signal V_{comp} .

In the embodiment considered, the sample-and-hold circuit **2140**, e.g. the electronic switch SW1, is controlled by a negative transient detection circuit **2142**. For example, this circuit may be configured to:

enable storage of the error signal V_{comp} (e.g. close the electronic switch SW1) when no negative load transient is detected; and

disable storage of the error signal V_{comp} (e.g. open the electronic switch SW1) when a negative load transient is detected.

For example, as described in the foregoing, the negative transient detection circuit **2142** may monitor for this purpose the variation or absolute value of the feedback signal FB, or the variation of the current sense signal CS. For example, such transients may be detected by determining the variation of a respective signal by comparing the signal with a previous value of the signal (e.g. stored via a sample-and-hold circuit, e.g. **2140**) or via a derivative network.

In the embodiment considered, the detector circuit **214** comprises a first current generator M1/M2/M3 configured to generate a current i_1 proportional to the stored error signal V_{comp} . Specifically, in the embodiments, the stored error signal V_{comp} is provided to a variable current generator M1, e.g. implemented with a FET, e.g. an n-channel FET, and a resistor Ra, configured to provide a current is proportional to the stored error signal V_{comp} . In various embodiments, the current is is provided also at input to a current mirror M2/M3, e.g. implemented with two FETs, such as p-channel FET, thereby providing at an output of the current mirror M2/M3 the current i_1 , which is applied to a node **2144**.

Similarly, in the embodiment considered, the detector circuit **214** comprises a second variable current generator M4, e.g. implemented with a FET, e.g. an n-channel FET, and a resistor Rb, configured to generate a current i_2 proportional to the current error signal V_{comp} .

Specifically, also the current generator M4 is connected to the node **2144**, whereby the node **2144** provides a current i_3 corresponding to the difference between the current i_1 and the current i_2 , i.e. $i_3 = i_1 - i_2$.

Accordingly, in the embodiment considered, the current i_3 corresponds to the control signal CTR being indicative of the difference between the stored and current value of the signal being indicative of the output current i_{out} .

For example, in the embodiment considered, the variable load **216** is implemented with a current mirror M5/M6, e.g. implemented with two FETs, such as n-channel FET, wherein the input (M5) of the current mirror receives the current i_3 and the output (M6) of the current mirror, which thus provides a current i_F proportional to the current i_3 , is connected between the terminals **202a** and **202b**.

Accordingly, by adjusting the gain of the various current generators and current mirrors, the proportionally between the current i_F and the variation of the output current i_{out} may be set, e.g. in order to reproduce (approximately) the same proportionally between the error signal V_{comp} and the average current I_L provided by the inductance L in response to

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the respective switching of the switches Q1 and Q2 generated via the blocks **208** and **210**.

FIG. 9 shows a second embodiment of the detector circuit **214** and the variable load **216**. Specifically, as mentioned before, the detector circuit **214** may be configured to reduce the signal CTR when the output load is static (step **1012**).

For example, in the embodiment considered, the detector circuit **214** comprises for this purpose a discharge circuit **2146** configured to selectively discharge the capacitor C_S . For example, in FIG. 9, the discharge circuit **2146** comprises an electronic switch SW2 and a current source **2148** connected in series between the output of the sample-and-hold circuit **2140**/input of the variable current generator M1/Ra and ground GND. For example, also the electronic switch SW2 may be driven via the control circuit **2142**.

In various embodiments, the detector circuit and/or the variable load **216** may be configured to be selectively enabled via an enable circuit **2150**. For example, in the embodiment considered, the enable circuit **2150** is implemented with an electronic switch S3 connected between the node **2144**, i.e. the output of the detector circuit **214**/the input of the variable load **216**, and ground GND. For example, also the electronic switch SW3 may be driven via the control circuit **2142**.

Accordingly, in various embodiments the control circuit **2142** may be configured to monitor whether a negative transient of the output current i_{out} occurs, and:

during a normal operation (in the absence of a negative transient, e.g. the negative variations is below a given threshold), close the electronic switch SW1 (i.e. enable storage of the sample-and-hold circuit **2140**) and the electronic switch SW3 (i.e. disable the variable load **216**), and open the electronic switch SW2 (i.e. progressive reduction of the control signal CTR is disabled), thereby following the output current and keeping off the variable load **216**, which in various embodiments corresponds to an active pull down;

once a negative transient current load is detected (e.g. a negative variation exceeds a given threshold), open the electronic switch SW1 (i.e. maintain the value stored by the sample-and-hold circuit **2140**) and open the electronic switch SW3 (i.e. enable the variable load **216**), whereby the current i_1 is linked to the stored output current and i_2 to the current output current, and the current i_F is proportional to the difference between these currents, i.e. $i_F = K(i_1 - i_2)$, where K represents a gain factor; and

once a new static condition is detected (in the absence of a transients, e.g. the variations are below a given threshold), close the electronic switch SW2, thereby reducing gradually the control signal CTR.

FIG. 10 shows an embodiment of an over-current protection circuit for the (high-side) electronic switch Q1. Specifically, as shown in FIG. 3, the current IQ, flowing through the electronic switch Q1 has a substantially linear behavior during the switch-on duration T_{ON1} , wherein the current increases from a minimum value at the instant t_1 to a maximum value at the instant t_2 . In the embodiment shown in FIG. 10, the circuit **50** is thus configured to generate a reference current i_R having a similar behavior, i.e. the circuit **50** is configured to generate a reference current i_R which increases linearly during the switch-on duration T_{ON1} from a minimum value at the instant t_1 to a maximum value at the instant t_2 , i.e. during the period when the signal DRV is set to high.

For example, in the embodiment considered, the circuit **50** comprises a ramp current generator **502**, **504** configured to

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generate a current ramp signal i_{Ramp} as a function of the pulsed signal DRV. For example, in the embodiment considered, the ramp current generator **502**, **504** comprises:

- a voltage ramp generator **502** configured to generate a ramp signal which is set to zero when the signal DRV is low, and then increased linearly when the signal DRV is high; and
- a variable current generator **504** configured to generate the current i_{Ramp} as a function of the voltage ramp signal provided by the voltage ramp generator **502**.

In various embodiments, the ramp current i_{Ramp} is provided to a node **520**.

Specifically, in various embodiments, the node **520** is also connected to a current generator **506** providing a constant offset current i_{Offset} .

In various embodiments, the node **520** is also connected to a further current generator **508** configured to determine a current i_c as a function of the signal V_{comp} , wherein the current i_c is preferably proportional to the signal V_{comp} . Specifically, in various embodiments, the current generator **508** uses a low-pass filtered version of the voltage V_{comp} as determined e.g. via a low-pass filter **508**.

Accordingly, in various embodiments, the current i_4 corresponds to:

$$i_4 = i_c - i_{Offset} - i_{Ramp}$$

Accordingly, in various embodiments, the current i_4 corresponds to a decreasing ramp signal.

In the embodiment considered, the current i_4 is then provided to a current limiter circuit **512** configured to generate the current i_R by limiting the current i_4 to a given maximum value i_{max} .

As shown in FIG. **10**, in various embodiments, the current i_4 may be provided to the current limiter circuit **512** indirectly by generating a current i_5 via a current mirror **M7**, **M8**, e.g. implemented with PMOS transistors.

For example, FIG. **11** shows an embodiment of the current limiter circuit **512**. Specifically, in the embodiment considered, the current limiter **512** is implemented with a Wilson current mirror, comprising three transistors, such as NMOS.

Specifically, in the embodiment considered, the Wilson current mirror comprises:

- a first branch comprising a transistor **M9** receiving the current i_4 or i_5 ;
- a second branch comprising two transistors **M11** and **M10** connected in series, wherein the gate terminal of the transistor **M11** is connected to the drain terminal of the transistor **M9**, the drain terminal of the transistor **M10** is connected to the source terminal of the transistor **M11**, and the gate and source terminals of the transistor **M10** are connected to the gate and source terminals of the transistor **M9**, respectively.

In various embodiments, a current limiter **518** may thus be connected in series with the second branch, thereby limiting the current flowing through the second branch to a maximum value i_{max} .

In the embodiment considered, a current mirror **M10**, **M12** may then be used to generate the current i_R by mirroring the current flowing through the second branch. Accordingly, in various embodiments, the current i_R corresponds to the current i_4 when $i_4 < i_{max}$ and i_{max} when $i_4 > i_{max}$.

In the embodiment considered, the reference current i_R is provided to a reference transistor **QR**, preferably corresponding to a scaled version of the transistor **Q1**. For example, in the embodiment considered, the reference transistor **QR** is an n-channel FET, e.g. a NMOS, wherein the drain terminal is connected to the terminal **200a**, the source

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terminal is connected to the reference current i_R , and the gate terminal is connected to the drive signal DRV₁. Specifically, by using a scaled version of the transistor **Q1**, also the reference current i_R may be a scaled version of the expected current profile of the current I_{Q1} .

Thus, by comparing the currents flowing through the transistors **Q1** and **QR** or the voltage at the transistors **Q1** and **QR**, the circuit **50** may detect whether the current I_{Q1} remains below the limit indicated by the reference current i_R . For example, in various embodiments, the voltage at the source terminals of the transistors **Q1** and **QR** are fed to a comparator **514** configured to generate a signal OC indicating that the current flowing through the transistor **Q1** exceeds the limit indicated by the current i_R . For example, in this case, the transistor **Q1** may be opened. For example, in the embodiment considered, the signal OC is fed via an OR gate (also receiving the signal provided by the comparator **2080**) to the reset input of the flip-flop **2082**.

Of course, without prejudice to the principle of the invention, the details of construction and the embodiments may vary widely with respect to what has been described and illustrated herein purely by way of example, without thereby departing from the scope of the present invention, as defined by the ensuing claims.

What is claimed is:

1. A method, comprising:

- storing a first signal indicative of an output current of a buck converter based on determining that a second signal indicative of a negative transient of the output current does not indicate the negative transient;
- generating a control signal as a function of a difference between the stored first signal and a third signal indicative of the output current, the generating the control signal being based on determining that a fourth signal indicative of the negative transient indicates the negative transient, the third signal and the fourth signal being subsequent signals in time to the first signal and the second signal; and

driving, by the buck converter, a variable load coupled between two output terminals of the buck converter as a function of the control signal.

2. The method of claim 1, wherein the determining that the second signal does not indicate the negative transient comprises one or more of:

- determining that the first signal indicates a decrease in the output current greater than a first value;
- determining that a feedback signal indicates an increase in an output voltage of the buck converter greater than a second value; and
- determining that the feedback signal indicates an increase in the output voltage greater than a threshold, or a combination thereof.

3. The method of claim 1, wherein the determining that the fourth signal indicates the negative transient comprises one or more of:

- determining that the third signal indicates a decrease in the output current greater than a first value;
- determining that a feedback signal indicates an increase in an output voltage of the buck converter greater than a second value; and
- determining that the feedback signal indicates an increase in the output voltage greater than a threshold, or a combination thereof.

4. The method of claim 1, further comprising reducing the control signal based on determining that the second signal or the fourth signal indicates a stable output current.

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5. The method of claim 4, wherein the reducing the control signal causes the variable load to draw less current.

6. The method of claim 1, further comprising:

generating an error signal as a function of a feedback signal indicative of an output voltage of the buck converter; and

generating a pulsed signal set to high for a first duration and set to low for a second duration, wherein the first duration and the second duration are varied as a function of the error signal.

7. The method of claim 6, further comprising generating a drive signal at a first terminal as a function of the pulsed signal, the first terminal being a terminal couplable to an electronic switch of the buck converter.

8. A circuit for a buck converter, the circuit configured to: store a first signal indicative of an output current of the buck converter based on determining that a second signal indicative of a negative transient of the output current does not indicate the negative transient; and generate a control signal as a function of a difference between the stored first signal and a third signal indicative of the output current, the generating the control signal being based on determining that a fourth signal indicative of the negative transient indicates the negative transient, the third signal and the fourth signal being subsequent signals in time to the first signal and the second signal, wherein the buck converter is configured to drive a variable load coupled between two output terminals of the buck converter as a function of the control signal.

9. The circuit of claim 8, further comprising a feedback terminal configured to receive a feedback signal indicative of an output voltage of the buck converter.

10. The circuit of claim 9, further comprising:

an error amplifier configured to generate an error signal as a function of the feedback signal;

a pulse generator circuit configured to generate a pulsed signal, the pulsed signal set high for a first duration and set low for a second duration, wherein the first duration and the second duration are varied as a function of the error signal; and

a drive circuit configured to generate a drive signal at a second terminal as a function of the pulsed signal, the second terminal configured to be coupled to an electronic switch of the buck converter.

11. The circuit of claim 9, further comprising an error amplifier configured to generate an error signal as a function of the feedback signal, the error amplifier being a regulator having an integral component, a proportional component, or an integral and a proportional component.

12. The circuit of claim 8, wherein the determining that the second signal does not indicate the negative transient comprises one or more of:

determining that the first signal indicates a decrease in the output current greater than a first value;

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determining that a feedback signal indicates an increase in an output voltage of the buck converter greater than a second value; and

determining that the feedback signal indicates an increase in the output voltage greater than a threshold.

13. The circuit of claim 8, wherein the determining that the fourth signal indicates the negative transient comprises one or more of:

determining that the third signal indicates a decrease in the output current greater than a first value;

determining that a feedback signal indicates an increase in an output voltage of the buck converter greater than a second value; and

determining that the feedback signal indicates an increase in the output voltage greater than a threshold, or a combination thereof.

14. A circuit of a buck converter, the circuit comprising: a sample and hold circuit configured to store, at a first time, a first signal indicative of an output current of the buck converter based on determining that a second signal indicative of a negative transient of the output current does not indicate the negative transient;

a first current generator configured to generate a first current proportional to the stored first signal at the first time; and

a second current generator configured to generate, at a second time, a second current proportional to a third signal indicative of the output current, wherein the circuit is configured to generate, at the second time, a control signal as a function of a difference between the first current and the second current, the generating the control signal being based on determining that a fourth signal indicative of the negative transient indicates the negative transient.

15. The circuit of claim 14, wherein the first current generator comprises a current mirror having a first field effect transistor (FET) and a second FET.

16. The circuit of claim 14, wherein the sample and hold circuit comprises a switch coupled to a capacitor.

17. The circuit of claim 14, wherein the second current generator comprises a field effect transistor (FET) coupled to a resistor.

18. The circuit of claim 14, further comprising a third current generator coupled to the first current generator, the third current generator configured to generate a third current proportional to the stored first signal at the first time, wherein the first current is a function of the third current.

19. The circuit of claim 14, wherein the first current generator is coupled to a voltage source.

20. The circuit of claim 14, wherein the buck converter is configured to drive a variable load coupled between two output terminals of the buck converter as a function of the control signal.

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